MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 7BSP7

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: 7-inch flat face, five beam, linear post accelerator, electrostatic focus and deflection.

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter Unit	Ef V	Ec1 Vdc	Eb1 Vdc	Eb2 Vdc	Eb3 Vdc	ed v	Rg Meg	Zd Meg (Note 1)	Ehk Vdc	Eb3/Eb2 Ratio
Maximum:	6.9	0	1,650	4,400	8,800	800	1.5	1.0	+180	2.0
Minimum:	5.7	-200	---	---	---	---	---	---	-180	---
TEST CONDITIONS (Note 2)	6.3	Adjust	Focus	3,000	6,000	---	---	1.0	---	2.0

GENERAL:

Qualification - Required (see note 3)

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
5248	Deflection - factor uniformity	-		---	---	?	?
---	Interaction factor	4		---	$6x10^{-5}$	---	in./Vdc
Ⓒ 1331	Direct - interelectrode capacitance	-	Cathode to all	Ck	---	6.0	pF
			Grid No. 1 to all	Cg1	---	6.0	pF
			D1 to D2	C1D2	---	3.0	pF
			D3 to D4	C3D4	---	2.0	pF
			D1 to all	CD1	---	4.5	pF
			D2 to all	CD2	---	4.5	pF
			D3 to all	CD3	---	4.0	pF
			D4 to all	CD4	---	4.0	pF
5111	Vibration	-		---	---	?	mm
1141	Pressure (implosion)	-		---	---	---	---
5216	Cathode illumination	-		---	---	---	---
	Quality conformance inspection, part 1						
5201	Electrode current (anode No.3)	5	Ec1 = Ec0	Ib3	---	18	µAdc
5201	Electrode current (anode No.3)	-	Ec1 = 0	Ib3	100	---	µAdc
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	9	Ib3 = 50 µAdc	---	---	---	---
5101	Angle between traces	-		---	88	92	Degrees
---	Trace alinement, corresponding traces	-		---	---	2	Degrees
5216	Stray light emission (conventional types)	-		---	---	---	---
Ⓒ 5106	Screen and faceplate blemishes	10		---	---	---	---
5223	Modulation	-	Ib3 = 25 µAdc	ΔEc1	---	35	Vdc
5226	Line width A	-	Ib3 = 2 µAdc	---	---	0.38	mm
5226	Line width B	-	Ib3 = 2 µAdc	---	---	0.89	mm
---	Line width A1	8	Ib3 = 2 µAdc	---	---	0.55	mm
5231	Spot position	-		---	---	20	mm
5231	Spot displacement (leakage)	-		---	---	10	mm
5241	Grid - cutoff voltage	-		Ec0	-42	-72	Vdc
5246	Focusing voltage	-	Ib3 = 25 µAdc	Eb1	755	1,000	Vdc
5103	Pattern distortion	6		---	---	---	---
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 1 leakage current	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2						
1301	Heater current	7		If	2,500	3,100	mAdc
5201	Electrode current (anode No.1)	-		Ib1	-15	+5	µAdc
5201	Electrode current (cathode)	-	Ib3 = 50 µAdc	Ic	---	6,000	µAdc
5101	Base alinement (electrostatic types)	-	+1D7, base key	---	---	---	---
5101	Side terminal alinement (electrostatic types)	-	+1D2	---	---	---	---
5101	Side terminal and base alinement	-	Base key	---	---	---	---
---	Collar pin No.1 and base key alinement	-		---	-10	+10	Degrees
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5221	Screen P7	-	Ib3 = 40 µAdc	---	---	---	---
5246	Focusing voltage	-		Fb1	---	1,050	Vdc
5246	Focusing voltage	-		Eb1	700	---	Vdc
5248	Deflection factor	-	1D2	DF	71	95	Vdc/in.
5248	Deflection factor	-	3D2	DF	77	105	Vdc/in.
5251	Heater - cathode leakage current	-		---	---	---	---
(C) 1101	Secureness of base, cap, or insert	-		---	---	---	---
(C) 1111	Base pin solder depth	-		---	---	---	---
(C) 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Ib3 = 10 µAdc, Eb3 = 8,800 Vdc; Eb2 = 4,400 Vdc; t = 500 hours	---	---	---	---
---	Life-test end points:						
5226	Line width A	-		---	---	0.56	mm
5226	Line width B	-		---	---	1.02	mm
5223	Modulation	-		ΔEc1	---	35	Vdc

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NOTES:

1. It is recommended that the deflecting-electrode circuit resistances be approximately equal. Higher resistance values up to 5.0 megohms may be used for low-beam current operation.

2. All tests except vibration, interaction factor and environmental tests shall be made on each unit separately. All readings of beam current (Ib3) shall be in addition to the reading obtained for anode No. 3 current at Ec1=Ec0. Each deflection plate shall be connected to A2 through a resistance of 1 megohm.

3. The activity responsible for the Qualified Products list is the Naval Electronic Systems Command, Department of Navy, Washington, D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification, (SD-6)". (Copies of "Provisions Governing Qualification (SD-6)" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

4. The deflection of one beam when balanced dc voltages are applied to deflection electrodes of the other beam will not be greater than the specified value.

5. With all beams cut off, the anode No. 3 current shall not exceed the limit specified.

6. All boundaries of a raster pattern, adjusted so its widest points just touch the sides of a 4.590-inch square, will fall within the area bounded by a 4.590-inch square and an inscribed 4.410-inch square.

7. Five heaters paralleled.

8. Measured in the center of the screen under the same conditions as for line width B.

9. This test is performed at the conclusion of the holding period.

© 10. Optical data.

a. Dead spots shall not exceed 0.025 inch in diameter nor exceed six in number. Separation of dead spots shall be 0.25 inch minimum.

b. No bright spots shall exceed 0.025 inch in diameter. Two spots up to 0.025 inch in diameter permitted. Six spots up to 0.020 inch in diameter permitted. Separation of bright spots shall be 0.25 inch minimum.

c. Color spots shall not exceed 0.025 inch in diameter nor exceed six in number. Separation of color spots shall be 0.25 inch minimum.

d. Elongated spots whose maximum width is 0.010 inch and whose maximum length is 0.25 inch shall be accepted.

e. The diameter of elongated spots whose width exceeds 0.010 inch shall be taken as half the sum of the length and width. The resultant diameter shall be assessed in accordance with a, b, and c, above.

f. Disregard all dead, bright, and color spots less than 0.010 inch. Separation shall be 0.25 inch minimum.

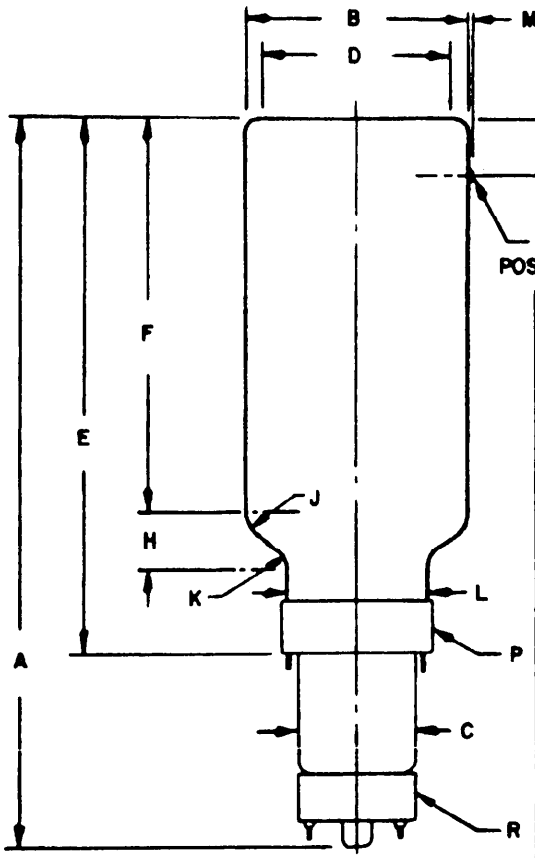
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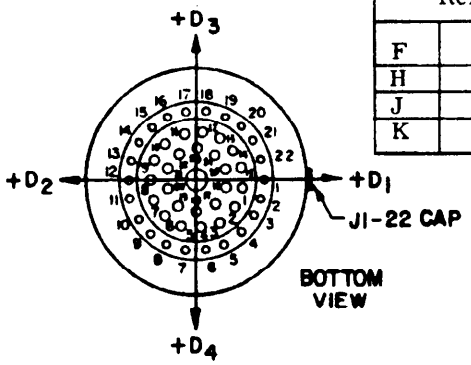
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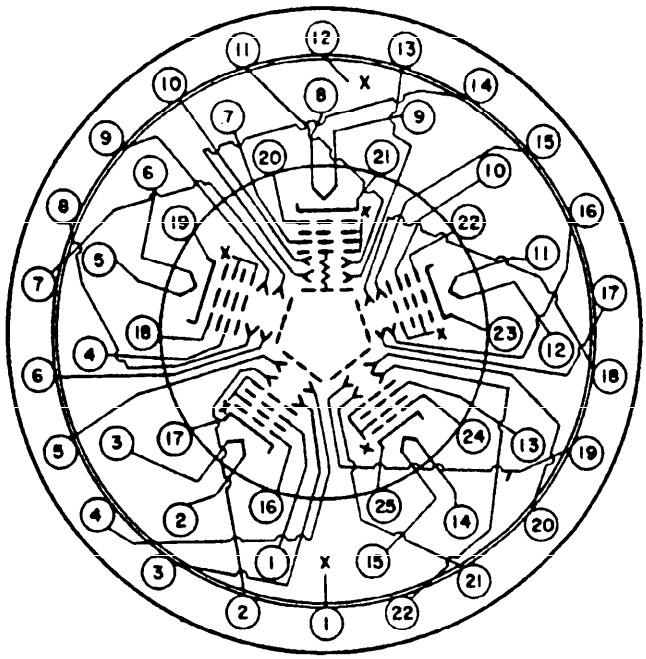
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
N	Bulb contact: J1-22	
P	Base, collar: 22-pin dimagnal	
R	Base: 25-pin pentaquintal, B25-139	
Quality conformance inspection, part 2		
A	22.625 (574.68)	23.125 (587.38)
B	6.875 (174.63)	7.125 (180.98)
C	3.688 (93.68)	3.812 (96.82)
D	6.000 (152.40)	---
E	16.312 (414.32)	17.063 (433.40)
G	1.625 (41.28)	1.875 (47.63)
L	4.312 (109.52)	4.500 (114.30)
M	---	.125 (3.18)
Reference dimensions (see note c)		
F	12.250 (311.15)	
H	1.750 (44.45)	
J	.750 (19.05) Rad.	
K	.750 (19.05) Rad.	



© NOTES:
a. Index pin, base key each alines with +1D2 trace ±10°.
b. J1-22 cap alines with +1D2 trace ±5°.
c. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 7BSP7.

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BASE & COLLAR CONNECTIONS

ELEMENT	UNIT "A"	UNIT "B"	UNIT "C"	UNIT "D"	UNIT "E"
	25-PIN BASE NUMBERS				
HEATER	2	5	8	11	14
HEATER	3	6	9	12	15
CATHODE	17	19	21	23	25
GRID NO.1	16	18	20	22	24
FOCUSING ELECTRODE	1	4	7	10	13
	22-PIN COLLAR NUMBERS				
ACCELERATOR (COMMON)	18&12	18&12	18&12	18&12	18&12
DEFLECTING ELECTRODE D_1	4	6	14	18	20
DEFLECTING ELECTRODE D_2	5	7	13	17	19
DEFLECTING ELECTRODE D_3	2	9	11	16	21
DEFLECTING ELECTRODE D_4	3	8	10	15	22

FIGURE 1. Outline drawing of electron tube type 7BSP7 - Continued.

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